STUDY REPORT (best version) — Persistent UAV-IoT Coverage Project

Read before reporting to senior professor. Dense but explanatory. Hinglish + simple English. Everything covered.

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0. One-line summary (pehle yeh padh, 30 sec)

UAV swarm ek area mein IoT sensors ko K-cover karta hai, battery finite, drones fail ho sakte hain, swarm connected rehna chahiye. Humne MAPPO (multi-agent RL) banaya, but ek hand-written rule ne usse 0.298 zyada coverage diya (gap). Paper ka real contribution = feasibility-certified evaluation framework jo batata hai ki koi config possible hai ya nahi, aur reported number meaningful hai ya nahi. Negative result honestly reported with controlled arms + 38-item audit + OMNeT negative control. Methodology contribution, not algorithm win.

1 PART A — OLD MIDSEM REPORT (kya tha, kya bolta tha, har section)

1.1 A.1 Old report ka framing (10 pages, proposal)

Old report title: "Persistent and Fault-Tolerant UAV-IoT Coverage: A Multi-Agent Reinforcement Learning Framework for Joint Coverage, Connectivity, and Energy Management." Authors same (Himanshu, Surya, Harsh, Gaddam). It was a **proposal**, not a results paper. Abstract says: "Numerical performance claims are intentionally deferred until the proposed experiments are completed." Contributions: (1) unified system model, (2) CTDE MAPPO framework with masks + reserve, (3) literature synthesis, (4) benchmark/metric protocol. No results, no feasibility check, no audit, no OMNeT execution. 10 pages, 5 figures, 14 references. Tone: optimistic, "we propose, we will do."

1.2 A.2 Old report ka system model (Section III) — har equation

Params (Table VI): N=5 active UAVs, 1 reserve, M=20 sensors, 80x80 m, Rcov=35 m, Rcomm=40 m, z0=50 m, K=2, dt=0.2s, vmax=5 m/s, dsafe=5 m, Bmax=30000 J, Phover=50 W, Pmove=75 W, Pelec=5 W, Ptx=0.5 W, charge increment 0.0002, T=10000 steps (2000 s).
Dec-POMDP: G=<N,S,Ai,Oi,T,R,gamma>. Global state st = (pi,vi,bi,mi,Fi) for all UAVs + sensors + Qt. Local obs oi = [pi, vi, bi, Ni, Clocal_i, rho_i, tau_return, Epad_i] (8 features). Action space: 6 actions (Hover, +/-x, +/-y, Charge). Joint action 6N. K-coverage: nd(t) = number of active UAVs covering sensor d. Ct = (1/M) sum min(nd/K,1). Ut = under-coverage rate. Connectivity: graph Gt, connected if one component. Articulation point indicator kappa_i. Crit = avg kappa. Battery: E(t+1) = clip(E - P*dt + Pch*dt). Return threshold Ereq = Epad + Erejoin + Ereserve. Mandatory return when E <= Ereq. Charging: Hungarian assignment, cost = alpha*dist + beta*queue + mu*urgency. Safety: speed, workspace, separation. Mask if violates. Failure/recovery: reserve UAV activated, recovery time tau_f. Reward: rt = -[lambdaO*Ut + lambdaE*Econs + lambdaD*Ddeg + lambdaW*Wmov + lambdaR*Rrec + lambdaC*Critt]. Weights: lambdaO=1.00, lambdaE=0.10, lambdaD=0.50, lambdaW=0.05, lambdaR=0.30, lambdaC=0.20. Feasibility (old): area counting N*pi*R*R >= K*A → Rcov >= sqrt(KA/piN) = 28.5 m; packing correction → 30.0 m. Both satisfied by 35 m. Surplus sigma = N*pi*R*R/(K*A) = 1.50. "Leaves margin for single-UAV failure (sigma=1.20 after failure)." Bound necessary not sufficient.

1.3 A.3 Old report ka ML algorithm (Section IV)

CTDE MAPPO: parameter-shared actor, centralized critic. GAE, clipped PPO, critic MSE loss. Action feasibility mask: workspace, speed, separation, charge validity, connectivity-preservation. Failure-aware execution: remove

failed UAV, activate reserve, update obs. Training protocol: randomize sensors, positions, batteries, pads, failure times; curriculum (no failures → single → contention). Hyperparameters: gamma=0.99, epsilon=0.20, GAE lambda=0.95. Implementation plan: Phase 1 Python (geometric, K-coverage, connectivity, battery, charging, failure, heuristic, MAPPO), Phase 2 OMNeT++/INET (wireless, network metrics), Phase 3 Simu5G (PHY-aware). Same actor interface across phases.

1.4 A.4 Old report ka comparison + metrics + scenarios (Sections V-VII)

Comparison models: Fixed formation+threshold recharge, Greedy lowest-battery, Reactive reconfiguration, MAD-DDQN, POGS-QMIX, MAHPPO-AM, Vanilla MAPPO. Metrics (Table IX): CK, under-coverage, connectivity availability, fragility, service continuity, outage severity, reserve utilization, recovery, energy, charging, movement, safety, scalability, stability. Scenarios S1-S8: sensor density, fleet size, battery, pads, Rcomm, Rcov, failure time/location, combined stress. Conclusion: defines problem, proposes MAPPO+mask+arbitration+reserve. "No performance advantage claimed before experiments." Literature survey: 10 core papers + 4 recent. Gap: no single framework couples all 5 elements.

1.5 A.5 Old report ka literature survey

10 core papers in Table I: [1] energy-efficient coverage DRL, [2] multi-ABS placement DDQN, [4] MAD-DDQN energy, [5] decentralized biconnectivity, [6] fault-tolerant ad hoc, [7] recharging coordination, [8] persistent aerial service, [9] fault-tolerant MARL exploration, [10] robust UAV coverage DRL, [14] energy-aware persistent coverage POGS-QMIX. Gap statement: no single framework models all 5 elements (K-coverage, connectivity criticality, battery+charging, failure recovery, decentralized execution). MAPPO cited separately [3].

2 PART B — What changed in new report and WHY

2.1 B.1 Config changes

- N: 5 → 8. WHY: feasibility certificate proves N=5 infeasible for region (N≥4K=8). Old area-surplus test was wrong.
- Episode: 10000 → 2000 steps. WHY: at 400 steps pack never depleted → charging/reserve never fired → three proposed mechanisms never exercised. Sweep shows 2000 steps is where energy binds (213 returns, 0.4 stranding/ep) while mission achievable.
- Initial deployment: unspecified → connected corner cluster (3/20 sensors). WHY: centred ring already covers 17/20 → CK≥0.85 for do-nothing controller; clustered makes comparison decidable. Pre-deployed kept as S9.

2.2 B.2 Reward changes

- 6 → 7 terms: added λB (Bt = mean fractional shortfall below return threshold). WHY: consumption-only signal gives no reason to turn back (flight to pad = 0.0016 of pack vs late team-level consequence).
- Weights: λD 0.50→2.00, λC 0.20→0.50. WHY: energy term numerically inert (≤5e-5 vs coverage 1); degraded-service and connectivity must not drown under coverage signal. New ordering: $\lambda D \geq \lambda O \geq \lambda C \geq \lambda R > \lambda E \geq \lambda W$.

2.3 B.3 Observation changes (8 → 11 features)

- Added one-hot identity 1i. WHY: parameter sharing needs symmetry breaking; without it, shared policy collapses to single action, swarm rigid body (measured: 1.84/8 distinct actions, spread 11.3m = cluster radius).
- Added pad bearing d-hat-pad. WHY: old obs carried energy cost to pad but not direction; policy cannot express return.
- Added G×G deficit summary Mdef (G=4). WHY: agent knows how many it covers, not where deficiency is. Measured: no coverage difference (null, reported) but behaviour changes (spread 9.0→10.6m).

2.4 B.4 Mask changes

- Generic connectivity mask → H=3 eccentricity lookahead + joint projection. WHY: individually legal moves can jointly disconnect; projection reverts all to previous config. Also analysis of what rule does/doesn't prevent (ring example: valid topology rejected). Measured: 0.10/16000 actions unexecuted.
- Energy mask repair: old admitted departures with flat battery → stranding 5.7/8. New requires E_i ≥ E_pad + E_reserve after the move, one-sided (already-low UAV may still move toward pad). Stranding 5.7→0.4.

2.5 B.5 New methodology (not in old report)

- Feasibility certificate (packing bound N≥4K=8). Old used area-surplus (wrong).
- Teleporting oracle (achievability ceiling 1.000 at N=8).
- Degeneracy theorem for SC (F=empty → SC=0 $\forall\pi$).
- Non-degenerate metrics (deficit + max refresh time).
- Claim audit (38 items, 0 fail).
- Determinism protocol (device + thread count part of protocol).
- Negative control for PHY (60 dB crush → same frame count → instrument fake).
- Plasticity probe (0% dormant vs 40-45%).
- Exhaustion vs failure distinction.

2.6 B.6 New implementation fixes (old equations got wrong in practice)

- Termination vs truncation: old treated end-of-mission as terminal → GAE bootstrapped from zero → biased advantages. Fix: separate flags, tests pin it.
- Value normalisation: old asserted present but missing. Fix: running EMA stats, un-normalise at 3 consumer points (TD residual, GAE, bootstrap). One accessor + tests.
- Motion step bug: first Phase-1 sim passed all equations/tests but motion step never applied moves → spread never changed. Found by behavioural diagnostic, not tests/loss.
- Frame count instrument: MAC frames responded to Ptx? No → 60 dB control caught it.

2.7 B.7 New literature positioning

- Old: 14 refs, 1 table, generic gap.
- New: 44+ refs, Table I (focused survey) + Table II (near-misses: Trotta [15], IRADA [44], Scherer/Rinner [16], Hartuv [17], Shi [18]) with exactly what each lacks.
- Dedicated subsection on closest prior work for evaluation philosophy ([38] drone-mesh ceilings, [39] CS-FUB, [40], [35]/[41] MILP).
- Bounded-absence claims: no infeasibility certificate for persistent coverage in corpus; no paper names degenerate binary metric. Stated as bounded search, not universal.
- Credit-assignment line: COMA, DAE, GPAE, HAPPO/HATRPO, wonderful-life-utility [26].

2.8 B.8 New results (none in old report)

- Base comparison: Fixed 0.372 / Greedy 0.784 / Reactive 0.953 / Reactive-local 0.972 / Indep DQN 0.381 / Masked PPO 0.625 / MAPPO 0.674±0.060 / argmax 0.418 / oracle 1.000.
- Gap 0.298 (CI [0.248,0.349]).
- Controlled arms (5) with CIs all below gap.
- Budget sweep 8/16/25 → 0.588/0.624/0.674.
- Component ablation (base + stress S8).
- Observation experiment (3 arms).
- Credit assignment test (-0.035).
- Scenario matrix (9 scenarios, 34/34 variants behind).
- Plasticity probe (0% dormant).

3 PART C — NEW REPORT (section by section, what it says)

3.1 C.1 Abstract (p.1)

UAV swarms provide temporary coverage where fixed infra unavailable. Persistent operation couples coverage, connectivity, battery, charging, failures. Formulates maintaining ≥K UAVs over every sensor while preserving connected swarm. Dec-POMDP. Argues evaluation methodology is the unsolved part: coverage ratio without achievability reference can't distinguish strong controller from infeasible config; service-continuity metric identically zero on infeasible instances. Introduces feasibility-certified framework: (i) rigorous lower bound on fleet size, (ii) teleporting placement oracle as ceiling, (iii) non-degenerate time-averaged + refresh-time metrics. Applied to project's own config: provably infeasible for region + minority of random sensor layouts; area-surplus certifies nothing. On corrected config: MAPPO with mask, reserve, arbitration, PHY-derived links. Hand-designed rule outperforms by 0.298 (CI [0.248,0.349]). Figure 1 is the argument in one picture. Arms bound how much each explanation accounts for (all upper limits below gap, smallest 0.161). Residual consistent with credit assignment; cheapest test (-0.035) doesn't resolve. Physical-model, feasibility, protocol, determinism claims recomputed by automated audit (38 items, 0 failing); audit doesn't cover Section VIII results. Every measurement number generated from artefact. Path-loss expression shown algebraically equivalent to independent standards-faithful simulator (constant 0.012 dB offset) with transmit-power negative control.

3.2 C.2 Introduction (pp.1-2)

Motivation (IoT in remote/disaster areas), persistent redundancy requirement, conflicts, joint action space 6N, MARL/CTDE/MAPPO suitability. Research question: can decentralized MAPPO maintain K-coverage + connectivity + energy + charging + failure recovery, and can performance be certified against rigorous achievability bound? 11 contributions listed (model, certificate, ceiling, negative result, MAPPO framework, PHY-derived links, non-degenerate protocol, observation-content spec, corrected deployments, negative-control methodology, literature synthesis).

3.3 C.3 Literature survey (pp.2-6)

Focused survey of 10 core papers (Table I) + near-misses (Table II: Trotta, IRADA, Scherer/Rinner, Hartuv, Shi, Qiu, Liu). Gap: no work couples all 5 elements with learned decentralized policy. Adjacent work on evaluation philosophy ([35],[41] MILP refs, [40] feasibility construct, [39] CS-FUB, [38] drone-mesh ceilings). Three absences: no infeasibility certificate for persistent coverage; no paper names degenerate binary metric; credit-assignment line (COMA, DAE, GPAE, HAPPO, wonderful-life-utility). Bounded search, not universal absence.

3.4 C.4 System model (pp.6-14)

Architecture (N active + 1 reserve + M sensors + sink + S pads). Dec-POMDP. Global state, local obs (11 features). Action space 6. K-coverage with radio gate (Eq. 19). Feasibility certificate (packing bound). Connectivity model (graph, articulation points). Radio model (Al-Hourani, path loss, MCS gate). Battery model (with exhaustion distinction). Charging arbitration. Safety constraints (energy mask one-sided). Failure/recovery (exhaustion vs failure). Primary objective + reward (7 terms, weights). Parameter settings (N=8, T=2000, clustered deployment). Scope (abstracts fading dynamics, obstacles, wind, aging, GPS denial, sensor failures, multi-failure).

3.5 C.5 ML algorithm (pp.14-18)

CTDE MAPPO (shared actor, centralized critic). Value normalisation (EMA, un-normalise at 3 points). Credit assignment analysis (shared team reward → same advantage; per-agent value doesn't repair; redistribution needs per-agent reward/joint-action critic/counterfactual baseline). Truncation vs termination bootstrap fix. Masking at logit level (H=3 eccentricity + joint projection, analysis of what it does/doesn't). Failure-aware execution. Training protocol. Hyperparameters. Implementation plan as audit trail (Phase 1-2 executed, Phase 3 scaffolded). Two cost confessions (motion step bug, frame-count instrument).

3.6 C.6 Comparison models (pp.18-19)

Fixed, Greedy, Reactive, Reactive-local, Independent DQN, Masked PPO local critic, POGS-QMIX (cited, not re-implemented), Vanilla MAPPO. "Evaluated" column. All on same scenes/seeds/constraints. Adaptations documented.

3.7 C.7 Metrics (pp.19-20)

CK, under-coverage, connectivity availability, fragility, SC (degenerate, retained for comparison), time-averaged deficit, max refresh time, outage severity, reserve utilization, recovery, energy, charging, movement, safety, scalability, stability. Which reported and which not (3 not reported: outage severity during recovery, recovery time distribution, inference-time scaling; collision/out-of-bounds prevented not measured).

3.8 C.8 Scenarios (pp.20)

S1-S9 (added S9 initial deployment). Each has two variants, every variant run. Evaluation protocol stated exactly (sampling seeded from eval seed; (policy,scene) function; seed-to-seed vs scene-to-scene variation reported separately; paired arms; resolution reported).

3.9 C.9 Results (pp.20-31)

Base comparison, gap 0.298, privilege test, SC separation, energy/stranding, connectivity accounting, budget sweep, observation experiment, energy arms, component ablation (base + stress), credit assignment, scenario matrix (34/34), claim audit, PHY validation (equivalence + negative control), plasticity probe.

3.10 C.10 Conclusion (pp.30-31)

Central contribution = feasibility-certified evaluation framework. Feasibility certificate ($N \geq 4K = 8$, tight, sensor-set probabilistic). What framework exposes (negative result 0.298, arms bound explanations, residual consistent with credit assignment). Scope of physical-layer claim (algebraic equivalence 0.012 dB, no runtime cross-validation). Honest limitations (ix). Remedies suggested (per-agent signal) not implemented.

3.11 C.11 References (pp.32-33)

44+ references including near-misses and credit-assignment line.

4 PART D — Glossary (34 terms)

Feasibility certificate
Proof that a config can K-cover region. Packing bound $N \geq 4K = 8$.

Teleporting oracle
Upper bound on coverage (drones placed anywhere, no motion/energy).

Service continuity (SC)
Binary metric, degenerate on infeasible configs.

Dec-POMDP
Cooperative multi-agent problem with partial obs.

CTDE MAPPO
Centralized training, decentralized execution, shared actor.

Feasibility mask
Safety filter: bounds, speed, separation, charge, connectivity.

Charging arbitration
Hungarian assignment of drones to pads.

Reserve UAV
Backup drone on failure.

Credit assignment
Shared reward → same advantage → who helped? Unresolved.

Stranding
UAV empties pack away from pad → removed from active set.

Exhaustion vs failure
Exhaustion = out of service, not failed; reserve can re-seat.

Radio gate
Coverage/adjacency only if link carries ≥ 6.5 Mbit/s.

Claim audit
38 items recomputed from artefacts, 0 fail.

OMNeT++/INET
Network simulator, Phase 2, 802.11 MAC/PHY.

BonnMotion
Mobility trace format (.movements).

Packing number P(A,R)
Max points in A pairwise $>2R$ apart.

Area surplus sigma
$N\pi R2/(KA)$; passes but certifies nothing.

GAE
Generalized advantage estimate.

PPO clipping
Limits policy update ratio.

H-step lookahead
Connectivity check H hops.

Joint projection
Reverts all moves if jointly disconnecting.

Entropy collapse
Policy loses exploration (ep 675 in long run).

BC init
Behaviour cloning init (97% acc) for residual MAPPO.

Residual MAPPO
BC init + PPO fine-tune (CK 0.819 after 3 ep).

Stress regimes
single_pad, small_battery, low_comm, failure, multi_failure.

Headroom gate
ceiling - best rule ≥ 0.10.

IQM
Interquartile mean (rliable stats).

NED
OMNeT network description language.

Cmdenv/Qtenv
OMNeT command-line / GUI run modes.

BreakpointPathLoss
INET path-loss model.

Al-Hourani
LoS probability model for air-to-ground.

MCS table
802.11 OFDM modulation/coding rates.

SNIR
Signal-to-noise-plus-interference ratio.

Stranded UAV
Out of service (empty pack away from pad).